UNIT-II

NEEDS FOR ELECTRICAL ENERGY STORAGE

Emerging Needs for EES

- Utilize more renewable Energy and Less Fossil Fuel

- The future Smart Grid uses

More Renewable Energy, less Fossil Fuel:

- **On- Grid Areas:**

Increased ratio of renewable generation may cause several issues in the power grid.

- Power fluctuation
- Undependability

Power fluctuation

Fluctuation in the output of renewable generation

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Makes system frequency control difficult

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Inefficient operation of thermal generators

Solution: EES can mitigate the output fluctuations

Undependability

Renewable energy output is undependable, since it is effected by weather conditions.

Measures to cope with this:

1. Increase the amount of renewable generation installed

2. Spread the installations of renewable generators over wide areas

- These measures are possible only with large number of installations and extension of transmission networks.

- Considering the cost of extra renewable generation and the difficulty of constructing new transmission facilities, **EES is a promising alternative measure**

Off- Grid Areas:

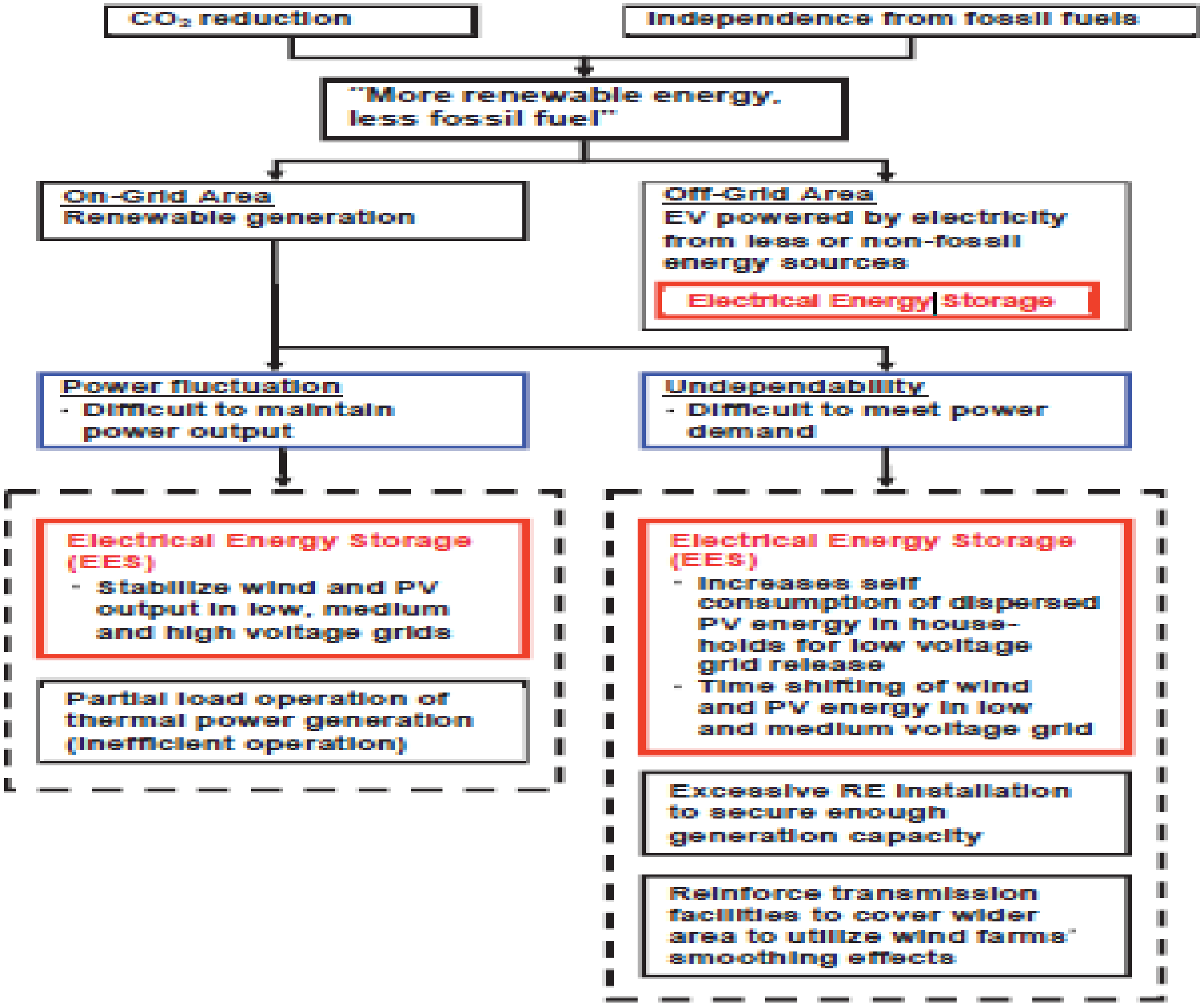
- Energy is consumed, particularly in the transport sector

- Fossil energy should be replaced with less or non-fossil energy

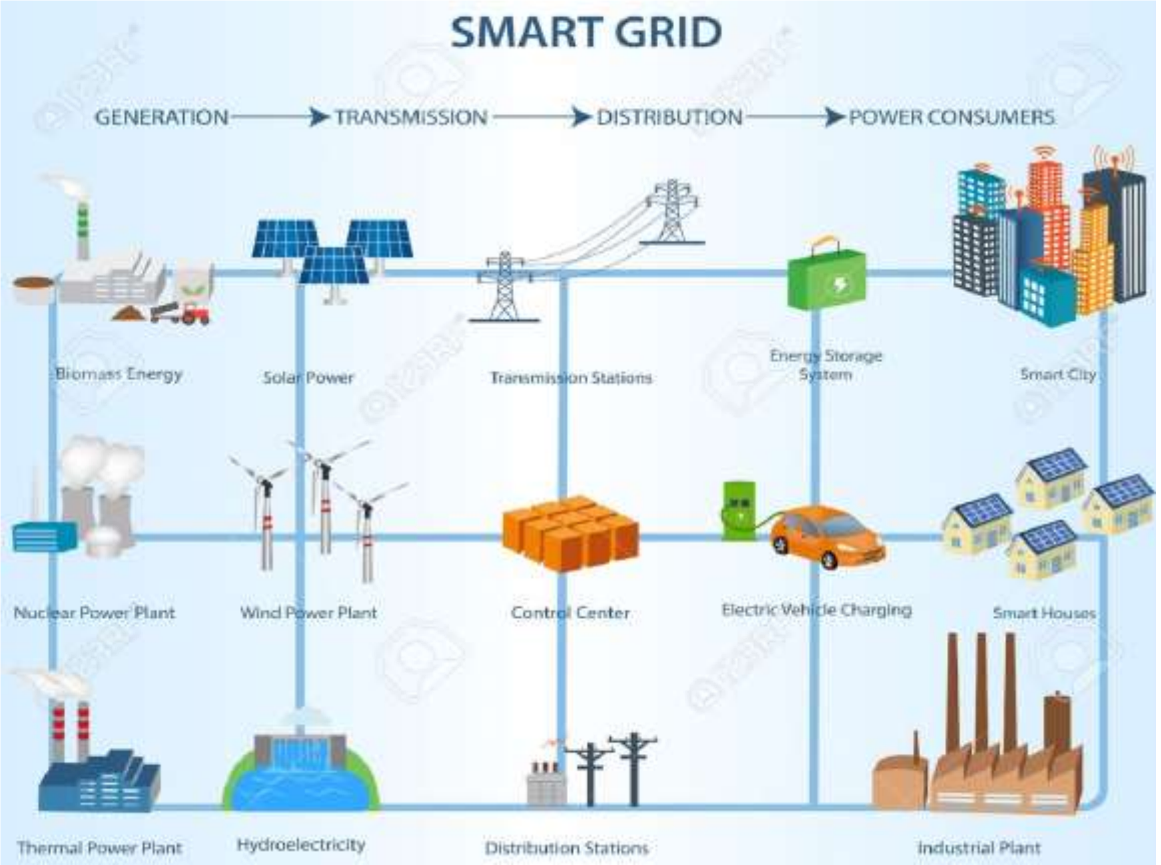
- Plug in Hybrid Electric Vehicles(PHEVs)

- Electric Vehicles

Problems in renewable energy installation and possible solutions



Smart Grid


SMART GRID
GENERATION
TRANSMISSION
DISTRIBUTION
POWER CONSUMERS
Biomass Energy
Solar Power
Transmission Stations
Energy Storage System
Smart City
Nuclear Power Plant
Wind Power Plant
Control Center
Electric Vehicle Charging
Smart Houses
Thermal Power Plant
Hydroelectricity
Distribution Stations
Industrial Plant


SMART GRID

Smart Grid

- The term Smart Grid is most commonly defined as an electric grid that has been digitized to enable two way communication between producers and consumers.

- The Smart Grid refers to the modernization of the electricity delivery system so that it monitors, protects and automatically optimizes the operation of its interconnected elements-from the central and distributed generator through the high-voltage network and distribution system, to industrial users and building automation systems, to energy storage installations and to end-use consumers including their electric vehicles, appliances and other household devices.

- The objective of the Smart Grid is to update electricity infrastructure to include more advanced communication, control, and sensory technology with the hope of increasing communication between consumers and energy producers.

- The potential benefits from a Smart Grid include
  - Increased reliability
  - More efficient electricity use
  - Better economics
  - Improved sustainability.

- A smart grid is an electrical grid which includes a variety of operational and energy measures including

- ✓ smart meters
- ✓ smart appliances
- ✓ renewable energy resources
- ✓ and energy efficient resources

- Electronic power conditioning and control of the production and distribution of electricity are important aspects of the smart grid.

- It represents an unprecedented opportunity to move the energy industry into a new era of

- Reliability

- Availability

- and efficiency

that will contribute to our economic and environmental health.

- <u>The benefits associated with the Smart Grid include</u>:

- ✓ More efficient transmission of electricity

- ✓ Quicker restoration of electricity after power disturbances

- ✓ Reduced operations and management costs for utilities, and ultimately lower power costs for consumers

- ✓ Reduced peak demand, which will also help lower electricity rates
- ✓ Increased integration of large-scale renewable energy systems
- ✓ Better integration of customer-owner power generation systems, including renewable energy systems
- ✓ Improved security

ESS in Smart Grid

- EES is expected to play an essential role in the future Smart Grid

- Some relevant applications:

1. EES installed in customer-side substations

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Can control power flow and mitigate congestion, or maintain voltage in the appropriate range

2. EES can support the electrification of existing equipment so as to integrate it into the smart grid EV's are → good examples

- Deployed in several regions

- Can serve as a mobile, distributed energy resource to provide a load shifting function in a smart grid

- Expected to be not only a new load for electricity but also a possible storage medium that could supply power to utilities when the electricity price is high

3. EES as the energy storage medium for Energy Management systems(EMS) in homes and buildings.

<u>Home EMS:</u>

A home energy management system is a technology platform comprised of both hardware and software that allows the user to monitor energy usage and production and to manually control and/or automate the use of energy within a household.

- With EMS, residential customers will become actively involved in modifying their energy spending patterns by monitoring their actual consumption in real time

- EMS in general will need EES

allowing EMS to function optimally with less power needed from the grid

Roles of Electrical Energy Storage Technologies

1. Roles from the viewpoint of a utility

2. Roles from the viewpoint of consumers

3.Roles from the viewpoint of generators of renewable energy

1. Roles from the view point of a utility

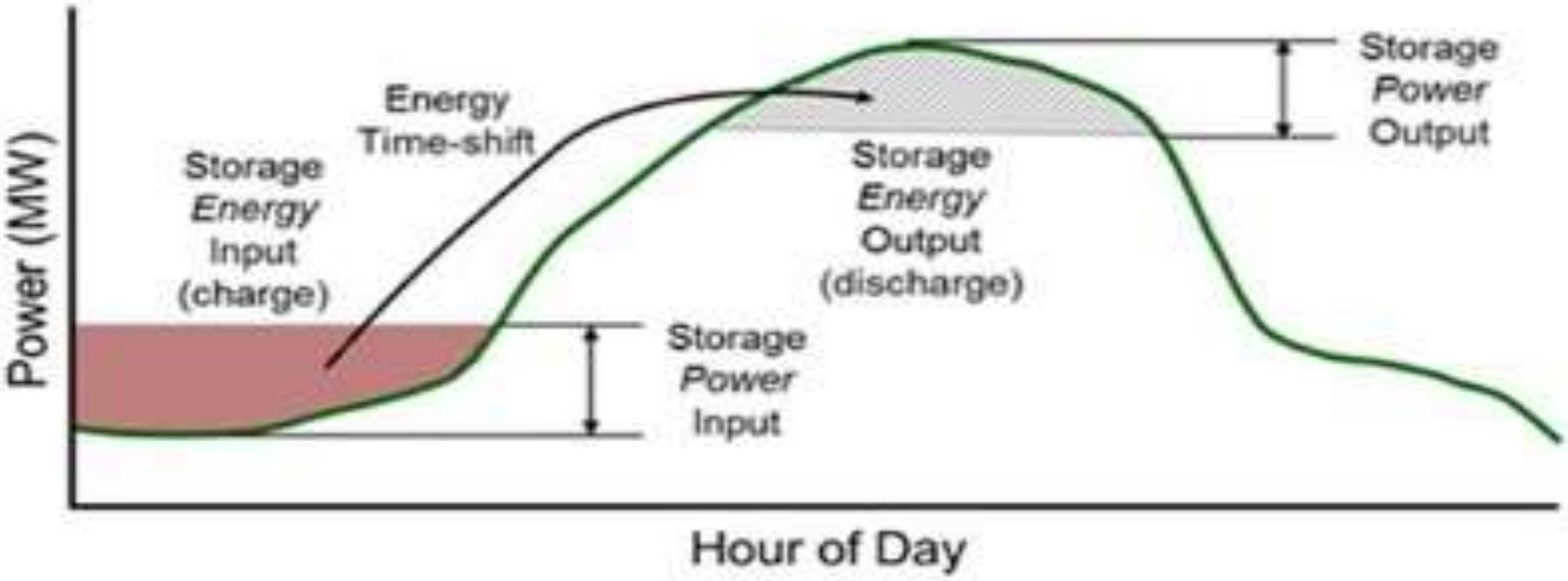
a. <u>Time shifting</u>:

- Utilities constantly need to prepare supply capacity and transmission/distribution lines

  to cope with annually increasing peak demand

- and has to consequently develop generation stations that produce electricity from primary energy.

- For utilities generation cost can be reduced by storing electricity at off-peak times and discharging it at peak times.



Source: E&I Consulting

Figure 1. Electric Energy Time-shift.

b. Power Quality:

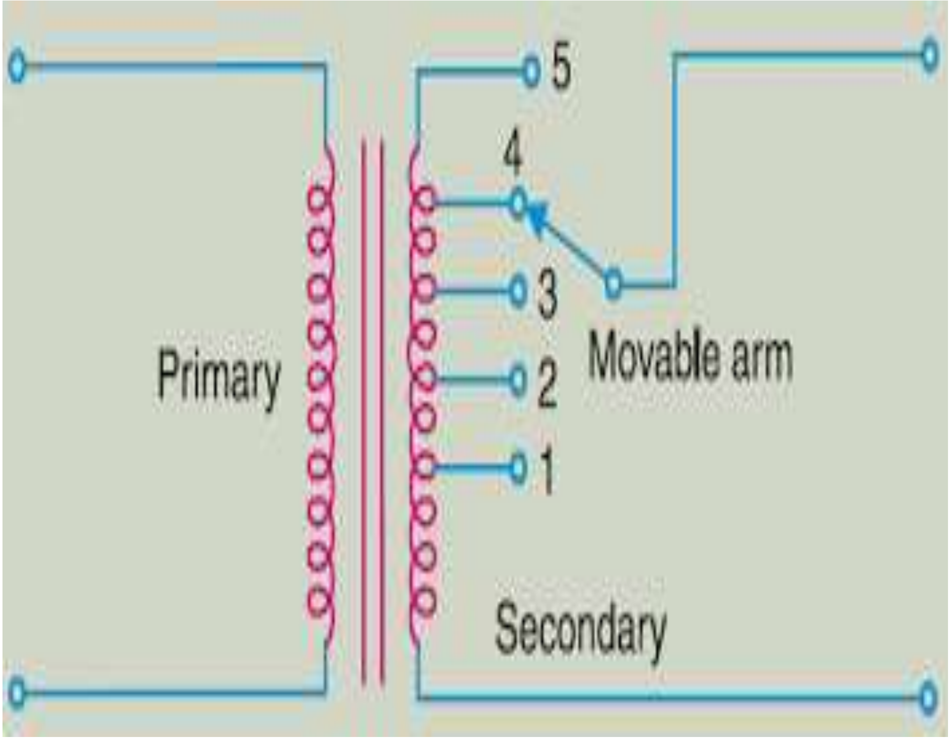
- A basic service that must be provided by utilities is to keep supply power voltage and frequency within tolerance, which they can do by adjusting supply to changing demand.

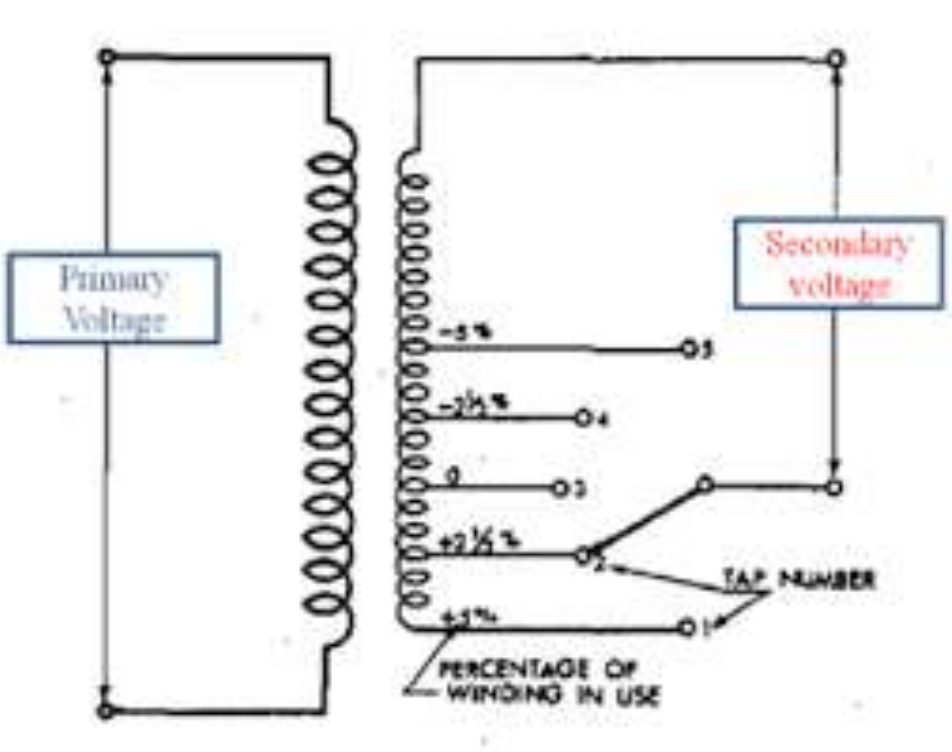
- Frequency control $\longrightarrow$ adjusting the output of power generators

$\updownarrow$

ESS can provide frequency control functions

- Voltage control ⟶ taps of transformers, & reactive power with phase modifiers





c. Making more efficient use of the network:

In a power network, congestion may occur when transmission/distribution lines cannot be reinforced in time to meet increasing power demand.

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Large scale batteries installed at appropriate substations may mitigate the congestion and thus help utilities to postpone or suspend the reinforcement of the network

d. Isolated grids:

- The power output from small-capacity generators such as diesel and renewable energy must match the power demand.

- ✓ By installing EES the utility can supply stable power to consumers.

e. Emergency power supply for protection and control equipment:

- A reliable power supply for protection and control is very important in power utilities

- ✓ Many batteries are used as an emergency power supply in case of outage

2. The roles from the view point of consumers

a. Time shifting/cost savings:

- Power utilities may set time-varying electricity prices(a lower price at night and higher price one during the day)

- ✓ Consumers may then reduce their electricity costs by using EES

b. Emergency Power supply :

- Consumers may possess appliances needing continuity of supply, such as fire sprinklers and security equipment.

- ✓ EES is sometimes installed as a substitute for emergency generators to operate during an outage

- Semiconductor and liquid-crystal manufacturers are greatly affected by even a momentary outage

- ESS technology such as large-scale batteries, double-layer capacitors and Super Conducting Magnetic Energy storage(SMES) can be installed

- A portable battery may also serve in an emergency to provide power to electrical appliances

c. Electric vehicles and mobile appliances:

- EVs are being promoted for CO_2 reduction

- ✓ High-performance batteries ⟹ nickel cadmium, nickel metal hydride and lithium ion batteries

- ✓ EV batteries are also expected to be used to power in-house appliances in combination with solar power and fuel cells

- Studies are being carried out to see whether they can usefully be connected to power networks

- ✓ V2H ⟶ vehicle to home

- ✓ V2G ⟶ vehicle to grid

3. The roles from the view point of generators of renewable energy

a. Time shifting:

- RE such as solar & wind power is subject to weather, and a surplus power may be thrown away when not needed on the demand

- ✓ Valuable energy can be effectively used by storing surplus electricity in EES and using it when necessary, it can also be sold when the price is high

b. Effective connection to grid:

- The output of solar & wind power generation greatly depending on the weather and wind speeds, which can make connecting them to the grid difficult.

- ✓ EES used for time shift can absorb this fluctuation more cost-effectively than other, single-purpose mitigation measures